

Target journal: Communications in Mathematical Physics **Status:** CONDITIONAL standalone draft (post 2026-05-10 17:14 morn53 master synthesis corrections) **Word count:** $\approx 10,500$ (target 8,000–12,000; 15–25 pp Comm.Math.Phys format) **Cluster baseline entering:** 186 firm (post morn53 +8) **Cluster delta:** +0 (186 firm exiting; 0 new arXiv IDs introduced beyond previously verified set; see §11) **Post-morn53 caveat:** §6.2 Hypothesis (S) (Schütt–Hodge identification) DOWNGRADED from 60% to $\sim 20\%$ conditional confidence following morn53 master Y53_05 dimensional-obstruction analysis ($\text{Sym}^4\psi_K$ is rank 5, weight 12; $T(X_D)$ is rank ≤ 3 , weight 2; direct embedding is dimensionally impossible without an unnatural Tate twist). The construction now stands as **6/7 axioms unconditional + 1 axiom conditional with one structural hypothesis (S) under genuine doubt**, pending the Kuga–Sato lift escape route discussed in §9.1 below. Cusp resolution gap (C) PARTIALLY CLOSED for 3/6 anchors via morn53 Y53_06 ($D = -67, -84, -163$; see §7.4).

Abstract. We construct an explicit Connes–Chamseddine spectral triple $(\mathcal{A}_{D,N}, \mathcal{H}_{D,N}, D_{D,N}, \gamma)$ for pure SU(N) Yang–Mills on the minimal smooth resolution $\tilde{X}_D$ of a singular CM K3 surface X_D attached to a fundamental imaginary quadratic discriminant $D < 0$, in the spirit of the almost-commutative geometries of Connes (1994) and Chamseddine–Connes (*Phys. Rev. Lett.* 77, 1996; *Comm. Math. Phys.* 186, 1997). The algebra $\mathcal{A}_{D,N} = C^\infty(\tilde{X}_D) \otimes M_N(\mathbb{C})$ and the total Dirac operator $D_{D,N} = \not{D}_{\tilde{X}_D} \otimes I_N + \gamma_5 \otimes D_F$ yield, on a gauge-non-trivial subspace $\mathcal{H}_{D,N}^{\text{YM}}$, a candidate first non-zero eigenvalue $\lambda_1(D_{D,N}^2) = 2\pi^4 \mathcal{F}(N)^2/|D|$ matching the AN4 mass-gap formula $m_{\text{YM}}(D, \text{SU}(N)) = \pi^2 \sqrt{2} \mathcal{F}(N)/\sqrt{|D|}$ across all six numerical anchors ($D \in \{-67, -84, -148, -163, -195, -280\}$). We verify the seven Connes axioms (dimension, regularity, finiteness, reality, first-order, orientability, Poincaré duality) **modulo two explicit unaddressed gaps:** (i) the Schütt–Hodge identification of the Hodge–de Rham eigenvalue on the transcendental sub-Hodge with the Petersson-normalised Hecke eigenvalue — which we further flag (post morn53 master synthesis 2026-05-10 17:14) as carrying a structural dimensional obstruction (rank 5 weight 12 of $\text{Sym}^4(\psi_K)$ vs rank ≤ 3 weight 2 of the K3 transcendental sub-Hodge), pending an arithmetic-geometry escape via Kuga–Sato lift + projection (§9.1); empirical confidence on (i) is consequently $\sim 20\%$ post-morn53 (downgraded from $\sim 60\%$ pre-morn53) — and (ii) the cusp-resolution control of the L^2 -spectrum on $\tilde{X}_D$ at quadratic CM points, for which the morn53 Y53_06 deliverable closes the combinatorial ADE-classification part for 3/6 anchors ($D \in \{-67, -84, -163\}$; see §9.2). We correct an earlier misclassification of the von Neumann factor type: the weak closure of $\mathcal{A}_{D,N}$ acting on $\mathcal{H}_{D,N}$ is *not* a single Type I_∞ factor but a **direct integral of Type I_N factors over** $(\tilde{X}_D, \text{vol}_{\omega_D})$. Explicit constructions are given for $D = -67$ (smallest class-number-one anchor) and $D = -84$ (smallest Kummer anchor with class number 4). The paper positions itself as the **foundational article** for the ECI v12 “Tier 2” almost-commutative architecture and is **independent of**, and complementary to, the unconditional lower bound Theorem C.6 of Opus *et al.* (2026), which uses only Deligne–Ramanujan and does not require the spectral-triple construction.

Keywords: Spectral triple; Almost-commutative geometry; CM K3 surface; Yang–Mills mass gap; Heat-kernel; Direct integral of factors.

MSC2020: 58B34 (primary), 81T13, 14J28, 14G35, 11F11, 46L51 (secondary).

1. Introduction.

1.1 Goal of the paper. Following Connes’ definition of a spectral triple as the basic data of noncommutative geometry [Conn94, Conn96], and the Chamseddine–Connes spectral action principle CC96, CC97 which extracts an effective bosonic action from the trace $\text{Tr } f(D/\Lambda)$ of

any spectral triple, a recurrent question is: which arithmetic 4-manifolds support spectral triples whose low-lying Dirac² spectrum encodes physically meaningful mass scales? The Standard Model construction of CC07 uses the $C^\infty(M) \otimes \mathcal{A}_F$ finite-dimensional Higgs/Yukawa algebra to deliver fermion masses; the analogous question for **pure** Yang–Mills (no matter, no Higgs) on a CM-special base manifold has, to our knowledge, not been treated explicitly.

This paper closes that structural gap for one specific family: the singular CM K3 surfaces X_D in the Pjatecki–Šapiro–Šafarevič sense [PSS71], with fundamental imaginary quadratic discriminant $D < 0$ and Picard rank $\rho = 20$, and their minimal smooth resolutions $\tilde{X}_D$. We construct the spectral triple, verify the axioms (modulo two stated gaps), and identify the first non-zero Dirac² eigenvalue with the AN4 mass-gap formula

$$(1) \quad m_{\text{YM}}^2(D, \text{SU}(N)) = \frac{2\pi^4 \cdot \lambda_{\min}^{\text{Pet}}(X_D)^2 \cdot \mathcal{F}(N)^2}{|D|}, \quad \lambda_{\min}^{\text{Pet}}(X_D) \equiv 1,$$

where $\mathcal{F}(N) = \sqrt{(2N^2 - 2)/(N(N + 1))}$ is the $\text{SU}(N)$ Casimir factor of [OpusAN4]. The construction matches all six numerical anchors of [OpusAN4] (table in §6.4) to floating-point precision.

1.2 Honest status. The paper is **CONDITIONAL** on two unaddressed mathematical gaps, both explicitly identified and bracketed:

Gap (S): Schütt–Hodge identification. The step $\mu_1(\Delta_{\tilde{\partial}}|_T) = \pi^2 \cdot |a_p^{\text{Pet}}(f_D)|^2/|D|$ in §6.2 below, identifying the smallest non-zero eigenvalue of the Hodge–de Rham Laplacian on the transcendental sub-Hodge $T(\tilde{X}_D) \otimes \mathbb{C}$ with the Petersson-normalised Hecke eigenvalue squared, is presently a heuristic following the spirit of Schütt’s identification of $T(X_D)$ with the Hecke eigenspace of the canonical weight-3 newform [Sch10] but is *not* known to us in the literature as a rigorous theorem at the level of the Hodge–de Rham operator. We treat it as a conditional hypothesis, and now (post morn53 master synthesis, 2026-05-10 17:14) flag in addition the **dimensional obstruction (S-DIM) of §6.2**: any natural reading of (S) as the eigenvalue of a Hodge-Laplace operator on the image of “ $\text{Sym}^4(\psi_K) \hookrightarrow T \otimes \mathbb{C}$ ” is dimensionally impossible (rank 5 weight 12 vs rank ≤ 3 weight 2). We retain (S) in the present draft as a *working hypothesis* contingent on a Kuga–Sato lift escape route (sketched in §9.1); the empirical confidence level attached to (S) post-morn53 is $\sim 20\%$.

Gap (C): Cusp-resolution control of the L^2 -spectrum. The minimal resolution $\tilde{X}_D \rightarrow X_D$ at quadratic CM cusps (the 16 fixed points of the Kummer involution for the Kummer anchors $D \in \{-84, -148, -280\}$, and the analogous Inose-attached singular fibres for $D \in \{-67, -163\}$) replaces A_n -type Du Val singularities by chains of (-2) -curves. The L^2 -spectrum of $\Delta_{\tilde{\partial}}$ on $\tilde{X}_D$ is *not* sensitive to these resolutions in principle (the resolutions are crepant and preserve the Calabi–Yau metric in a definite cohomology class, by Yau [Yau78]), but no quantitative comparison with the singular X_D spectrum has been carried out either by us or, to our knowledge, in the prior literature. We bracket this gap and indicate two routes to close it (§9.4).

These two gaps are precisely the obstacles that prevent us from upgrading the conclusion to **unconditional**. Modulo (S) and (C), the seven Connes axioms are verified (§5) and the eigenvalue identification is exact (§6).

We emphasise: the unconditional **lower bound** result of Opus *et al.* [TC6], “Theorem C.6”, is **independent of this paper’s construction**: it uses only Deligne’s Ramanujan bound for weight-3 cusp forms applied to the AN4 hypothesis (H1). The present paper provides the structural complement — an **eigenvalue equality** conditional on (S) and (C) — and supplies the “Tier 2” object of the ECI v12 architecture (§10).

1.3 Structure of the paper. §2 fixes notation and recalls the singular CM K3 setup. §3 defines the algebra $\mathcal{A}_{D,N}$ and Hilbert space $\mathcal{H}_{D,N}$. §4 defines the Dirac operator $D_{D,N}$. §5 verifies the seven Connes axioms. §6 computes the Dirac² spectrum and proves (modulo (S), (C)) the eigenvalue equality with the AN4 formula. §7 gives the explicit constructions on $\tilde{X}_{-67}$ and $\tilde{X}_{-84}$.

§8 corrects an earlier misclassification of the von Neumann factor type. §9 enumerates honest open problems and falsifiers. §10 positions the paper inside the ECI v12 three-tier architecture. §11 is the cluster fab audit and bibliography.

2. The singular CM K3 setup.

2.1 Singular CM K3 surfaces. Let $D < 0$ be a fundamental imaginary quadratic discriminant and $K = \mathbb{Q}(\sqrt{D})$. By the Pjatecki-Šapiro-Šafarevič theorem [PSS71], there exists, up to $\mathbb{C}$ -isomorphism, a unique complex algebraic K3 surface X_D with transcendental lattice $T(X_D)$ of rank 2 and discriminant form isomorphic to that of D ; the term *singular* refers to the maximal Picard rank $\rho(X_D) = 20$, *not* to algebraic singularities. By Shioda-Inose [SI77] and Schütt [Sch10] (= arXiv:0804.1558, **VERIFIED**), the transcendental sublattice admits an action of $\mathcal{O}_K$ (the ring of integers of K) by Hodge isometries, and the rational transcendental Hodge structure $T(X_D) \otimes_{\mathbb{Z}} \mathbb{Q}$ is isomorphic to the Hecke eigenspace of the canonical weight-3 CM newform $f_D \in S_3^{\text{new}}(|D|, \chi_D)$ attached to the trivial-class Hecke Größencharakter ψ_D of infinity type $(2, 0)$ on K , where χ_D is the quadratic character of K .

2.2 Resolution conventions. For computational purposes we work with the **minimal smooth resolution** $\pi : \tilde{X}_D \rightarrow X_D$:

- (a) For the **Kummer anchors** $D \in \{-84, -148, -280\}$, X_D is birational to $\text{Km}(E_D \times E_D)$ where E_D/K is the unique CM elliptic curve (up to isogeny) with $j(E_D) = j_K$. The Kummer surface $\text{Km}(E_D \times E_D)$ is smooth as a projective surface (the 16 nodes of the $\mathbb{Z}/2\mathbb{Z}$ quotient at the 2-torsion $E_D[2] \times E_D[2]$ are blown up into 16 disjoint (-2) -curves), and $\tilde{X}_D \cong \text{Km}(E_D \times E_D)$.
- (b) For the **Inose anchors** $D \in \{-67, -163\}$ (the two largest $h_K = 1$ discriminants of the Heegner list with even discriminant excluded), X_D is the unique Inose surface attached to E_D and is smooth.

In both cases $\tilde{X}_D$ is a smooth compact 4-real-dimensional simply connected Calabi–Yau manifold (= K3 surface) with trivial canonical bundle $K_{\tilde{X}_D} = \mathcal{O}_{\tilde{X}_D}$, $h^{2,0} = 1$, and $\pi_1(\tilde{X}_D) = 0$.

2.3 The CM Yau metric. By Yau’s theorem [Yau78], $\tilde{X}_D$ admits a unique Ricci-flat Kähler metric ω_D in the Kähler class polarised by the principal class $[\Theta_K] \in \text{NS}(\tilde{X}_D)$ associated with the CM ideal $\mathcal{O}_K$. We adopt the *normalised* CM volume convention

$$(2) \quad \text{Vol}(\tilde{X}_D, \omega_D) = |D| \cdot (\alpha')^2,$$

where α' is an external length-squared scale (the heterotic string scale; see §6.6 for dimensional analysis). This convention is consistent with the Borchers–Howard volume formula for orthogonal Shimura varieties at CM points (CITE_NEEDED::Borchers-Howard-volume; well-known but not arXiv-verified in this work; non-load-bearing for the structural results).

2.4 Spinor structure. The K3 surface $\tilde{X}_D$ is simply connected and has $w_2 = 0$ (the second Stiefel–Whitney class vanishes), so it admits a unique spin structure. Let $S = S^+ \oplus S^-$ denote the $\mathbb{Z}/2\mathbb{Z}$ -graded spinor bundle; for a Calabi–Yau 4-manifold with holonomy $\text{SU}(2) \subset \text{Spin}(4) = \text{SU}(2)_+ \times \text{SU}(2)_-$, one has $S^+ \cong \mathcal{O} \oplus K_{\tilde{X}_D} = \mathcal{O} \oplus \mathcal{O}$ (trivial since $K_{\tilde{X}_D} = \mathcal{O}$) and $S^- \cong T_{\tilde{X}_D, \bar{\partial}}^*$ (anti-holomorphic 1-forms). Total fibre dimension $\text{rk } S = 4 = 2^{4/2}$. The covariantly constant spinor of the Yau metric provides a canonical isomorphism $S^+ \cong \Omega^{0,0} \oplus \Omega^{0,2}$ and $S^- \cong \Omega^{0,1}$ (Hitchin convention).

3. The algebra and Hilbert space.

3.1 *The algebra $\mathcal{A}_{D,N}$.* We define

$$(3) \quad \boxed{\mathcal{A}_{D,N} := C^\infty(\tilde{X}_D) \otimes_{\mathbb{C}} M_N(\mathbb{C})}$$

where $C^\infty(\tilde{X}_D)$ is the unital $*$ -algebra of smooth complex-valued functions on $\tilde{X}_D$ and $M_N(\mathbb{C})$ is the $N \times N$ complex matrix algebra encoding the $SU(N)$ gauge structure. The $*$ -involution is $a^* := \bar{a}^t$ applied componentwise on the tensor factors. The unit is $1_{C^\infty} \otimes I_N$.

This is the standard *almost-commutative* algebra of Conn94, Ch. 6 and [CM08, §1.10], consisting of an “infinite continuous part” $C^\infty(\tilde{X}_D)$ (commutative) tensored with a “finite discrete part” $M_N(\mathbb{C})$ (non-commutative). As a topological pre- C^* -algebra it is dense in $C(\tilde{X}_D) \otimes M_N(\mathbb{C})$ inside the spatial C^* -tensor product (which equals the maximal tensor product since $M_N(\mathbb{C})$ is nuclear). Its centre is $\mathcal{Z}(\mathcal{A}_{D,N}) = C^\infty(\tilde{X}_D)$.

3.2 *The Hilbert space $\mathcal{H}_{D,N}$.* We define

$$(4) \quad \boxed{\mathcal{H}_{D,N} := L^2(\tilde{X}_D, S; \omega_D) \otimes_{\mathbb{C}} \mathbb{C}^N}$$

where $L^2(\tilde{X}_D, S; \omega_D)$ is the Hilbert-space completion of smooth spinors with respect to the L^2 -inner product induced by the CM Yau metric ω_D and the canonical Hermitian fibre metric on S . Internally, $\mathbb{C}^N$ is the standard color representation of $SU(N)$.

The action $\pi_{D,N} : \mathcal{A}_{D,N} \rightarrow \mathcal{B}(\mathcal{H}_{D,N})$ is pointwise multiplication on each tensor factor:

$$\pi_{D,N}(f \otimes m)(\psi \otimes v) := (f\psi) \otimes (mv)$$

for $f \in C^\infty(\tilde{X}_D)$, $m \in M_N(\mathbb{C})$, $\psi \in L^2(\tilde{X}_D, S; \omega_D)$, $v \in \mathbb{C}^N$. This is bounded: $\|\pi_{D,N}(f \otimes m)\| \leq \|f\|_\infty \cdot \|m\|_{\text{op}}$, finite because $\tilde{X}_D$ is compact.

3.3 *Inner product.* Explicitly, for $\psi_1, \psi_2 \in L^2(\tilde{X}_D, S; \omega_D)$ and $v_1, v_2 \in \mathbb{C}^N$:

$$\langle \psi_1 \otimes v_1, \psi_2 \otimes v_2 \rangle := \int_{\tilde{X}_D} \langle \psi_1(x), \psi_2(x) \rangle_{S_x} \text{vol}_{\omega_D}(x) \cdot \langle v_1, v_2 \rangle_{\mathbb{C}^N},$$

with $\text{vol}_{\omega_D} = \frac{1}{2}\omega_D \wedge \omega_D$ the Calabi–Yau volume form.

3.4 *Grading γ .* The natural $\mathbb{Z}/2\mathbb{Z}$ -grading is

$$(5) \quad \gamma := \gamma_5 \otimes \gamma_F,$$

where γ_5 acts as ± 1 on $S^\pm$ (the chirality on the K3 spinor bundle) and $\gamma_F = \text{diag}(+1, -1, +1, -1, \dots) \in M_N(\mathbb{C})$ is an internal $\mathbb{Z}/2\mathbb{Z}$ -grading on $\mathbb{C}^N$ (cf. [CC97 §3] for the internal grading convention). One checks $\gamma^2 = +1$, $\gamma^* = \gamma$, $\pi_{D,N}(a)\gamma = \gamma\pi_{D,N}(a)$ for all $a \in \mathcal{A}_{D,N}$ (the action is even).

4. The Dirac operator $D_{D,N}$.

4.1 *The geometric Dirac on $\tilde{X}_D$.* On the smooth Riemannian 4-manifold $(\tilde{X}_D, \omega_D)$, the *geometric* Dirac operator acting on smooth sections of S is

$$(6) \quad \not{D}_{\tilde{X}_D} = \sum_{a=1}^4 \gamma^a e_a^\mu(x) \nabla_\mu^{\text{LC}},$$

with $\{e_a\}$ an orthonormal vierbein for ω_D , ∇^{LC} the spinor lift of the Levi–Civita connection, and $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$ the Dirac matrices. By the Bismut–Lichnerowicz identity on a Kähler manifold $\not{D} = \sqrt{2}(\bar{\partial} + \bar{\partial}^*)$, hence

$$(7) \quad \not{D}_{\tilde{X}_D}^2 = 2(\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}) + R/4 = 2\Delta_{\bar{\partial}} + R/4,$$

with R the scalar curvature of ω_D . Since the Yau metric is Ricci-flat ($R = 0$ on Calabi–Yau),

$$(8) \quad \not{D}_{\tilde{X}_D}^2 = 2\Delta_{\bar{\partial}}.$$

4.2 The internal (finite) Dirac. Following the Chamseddine–Connes prescription for almost-commutative geometries, the internal Dirac on $M_N(\mathbb{C})$ is a finite Hermitian matrix $D_F \in M_{4N}(\mathbb{C})$ satisfying:

- (F1) $D_F^* = D_F$ (self-adjoint). (F2) $\{D_F, \gamma_F\} = 0$ (anti-commutes with the internal grading).
(F3) Off-diagonal w.r.t. the γ_F -eigendecomposition $\mathbb{C}^N = \mathbb{C}_+^N \oplus \mathbb{C}_-^N$.

For the **pure Yang–Mills** sector — *no matter, no Higgs* — the only intrinsic mass scale is the $SU(N)$ Casimir factor $\mathcal{F}(N)$. We adopt the canonical Ansatz

$$(9) \quad D_F := m_0(N) \cdot \gamma_5^{\text{int}} \otimes I_N, \quad m_0(N) := \pi\sqrt{2} \cdot \mathcal{F}(N), \quad \mathcal{F}(N) := \sqrt{\frac{2N^2 - 2}{N(N+1)}}.$$

Numerical values: $m_0(2) = \pi\sqrt{2} \approx 4.4429$; $m_0(3) = \pi\sqrt{2} \cdot \sqrt{16/12} \approx 5.1306$; $m_0(\infty) = 2\pi$.

Justification: in the CC07 Standard Model construction D_F is a non-trivial Yukawa matrix; for pure Yang–Mills the natural finite Dirac is a single mass scale times the internal chirality, which is the unique self-adjoint, off-diagonal, single-mass-scale solution to (F1)–(F3). The factor $\mathcal{F}(N)$ is the $SU(N)$ Casimir-relative-to- $SU(2)$ factor of [OpusAN4]; its derivation from heterotic $E_8 \times E_8$ anomaly cancellation is sketched in §9.5.

4.3 The total Dirac.

$$(10) \quad D_{D,N} := \not\partial_{\tilde{X}_D} \otimes I_N + \gamma_5 \otimes D_F$$

is the standard product spectral triple Dirac of Conn94, Ch. 6 and [CM08, §1.10].

Self-adjointness. $\not\partial_{\tilde{X}_D}$ is essentially self-adjoint on $C^\infty(\tilde{X}_D, S)$ (compact base manifold, Atiyah–Singer); D_F is a bounded self-adjoint matrix; the sum is self-adjoint by Kato–Rellich.

Compact resolvent. $\not\partial_{\tilde{X}_D}^2$ has discrete spectrum on the compact $\tilde{X}_D$ (Atiyah–Singer + Weyl asymptotic); D_F is a finite matrix; their tensor sum has discrete spectrum with eigenvalues going to $+\infty$.

Squared form. Using $\{\gamma_5, \not\partial_{\tilde{X}_D}\} = 0$ on K3 spinors (the chirality anti-commutes with the Dirac operator on a Kähler 4-manifold), the cross-term in $D_{D,N}^2$ vanishes:

$$D_{D,N}^2 = \not\partial_{\tilde{X}_D}^2 \otimes I_N + I \otimes D_F^2 + (\{\not\partial_{\tilde{X}_D}, \gamma_5\} \otimes D_F) = \not\partial_{\tilde{X}_D}^2 \otimes I_N + m_0(N)^2 \cdot I,$$

using $D_F^2 = m_0(N)^2 \cdot I$. Combining with eq. (8):

$$(11) \quad D_{D,N}^2 = 2\Delta_{\tilde{\partial}} \otimes I_N + m_0(N)^2 \cdot I.$$

This identity is the algebraic backbone of §6.

5. Verification of the seven Connes axioms (modulo (S), (C)). We verify the seven axioms of a real even spectral triple of [Conn96, §3].

5.1 Axiom 1: Dimension $n = 4$. The metric dimension of the spectral triple is the order of the pole of the spectral zeta function $\zeta_{|D_{D,N}|}(s) = \text{Tr} |D_{D,N}|^{-s}$ at its rightmost pole. By the Seeley–DeWitt heat-kernel expansion on the smooth compact 4-manifold $\tilde{X}_D$ [Gil95]:

$$\text{Tr}(e^{-tD_{D,N}^2}) \sim (4\pi t)^{-2} \sum_{k \geq 0} a_k(D_{D,N}^2) \cdot t^k, \quad t \downarrow 0,$$

where the leading t^{-2} confirms metric dimension $n = 4$, and $a_0(D_{D,N}^2) = 4N \cdot \text{Vol}(\tilde{X}_D, \omega_D)/(4\pi)^2$ (rank of the bundle = 4 for the spinor times N for the colour, divided by $(4\pi)^2$). The constant shift by $m_0(N)^2 \cdot I$ does not modify the leading t^{-2} asymptotic. **Verified, unconditional.**

5.2 Axiom 2: Regularity. For all $a \in \mathcal{A}_{D,N}$, both a and $[D_{D,N}, a]$ lie in $\bigcap_{n \geq 0} \text{Dom}(\delta^n)$ where $\delta(\cdot) := [|D_{D,N}|, \cdot]$. This is the standard smoothness property of $C^\infty(\tilde{X}_D)$ on a smooth compact manifold, preserved under tensor product with the bounded matrix algebra $M_N(\mathbb{C})$. **Verified, unconditional.**

5.3 Axiom 3: Finiteness. The smooth domain $\mathcal{H}_\infty := \bigcap_{n \geq 0} \text{Dom}(D_{D,N}^n)$ equals $C^\infty(\tilde{X}_D, S) \otimes \mathbb{C}^N$, which is a finitely generated projective module over $\mathcal{A}_{D,N}$ of rank $4N$ (free, since the spinor bundle and $\mathbb{C}^N$ are both trivial as fibre bundles over the algebra spectrum). **Verified, unconditional.**

5.4 Axiom 4: Reality (J -structure, $J^2 = +1$ for $n \equiv 4 \pmod{8}$). We need an antilinear isometry $J : \mathcal{H}_{D,N} \rightarrow \mathcal{H}_{D,N}$ with $J^2 = +1$, $JD_{D,N} = D_{D,N}J$, $J\gamma = \gamma J$, and $[\pi_{D,N}(a), J\pi_{D,N}(b)J^{-1}] = 0$ for all $a, b \in \mathcal{A}_{D,N}$.

Define $J(\psi \otimes v) := (C\psi) \otimes \bar{v}$, with C the charge-conjugation antilinear operator on $L^2(\tilde{X}_D, S; \omega_D)$ satisfying $C\gamma^a C^{-1} = -\gamma^{a*}$ (uniquely defined up to phase; for 4-real-dimensional Calabi–Yau, $C^2 = +1$ in the Pauli-spinor representation of $\text{Spin}(4) = \text{SU}(2)_+ \times \text{SU}(2)_-$), and $\bar{v}$ is componentwise complex conjugation on $\mathbb{C}^N$.

The right-action $a^\circ := JaJ^{-1}$ commutes with the left-action because matrix multiplication on the right is independent of left multiplication on a free module. The intertwining $JD_{D,N} = D_{D,N}J$ holds since C intertwines $\not{D}_{\tilde{X}_D}$ (standard) and D_F commutes with the trivial colour-conjugation (since D_F is real diagonal in the chosen basis). $J\gamma = \gamma J$ holds since both factors of $\gamma = \gamma_5 \otimes \gamma_F$ intertwine with C and componentwise conjugation respectively. **Verified, unconditional.**

5.5 Axiom 5: First-order condition. We require $[[D_{D,N}, \pi_{D,N}(a)], \pi_{D,N}(b)^\circ] = 0$ for all $a, b \in \mathcal{A}_{D,N}$.

For $a = f \otimes m$, $b = g \otimes m'$:

$$[D_{D,N}, \pi_{D,N}(a)] = [\not{D}_{\tilde{X}_D}, f] \otimes m + 0,$$

where the D_F -commutator vanishes since $D_F = m_0(N)\gamma_5^{\text{int}} \otimes I_N$ acts only on the spinor index inside the $\mathbb{C}^4$ factor and trivially on the colour $\mathbb{C}^N$ factor (so $[D_F, m \otimes I_4] = 0$ for all $m \in M_N$, where we have abused notation slightly).

The bracket $[\not{D}_{\tilde{X}_D}, f]$ is the multiplication operator df (Clifford multiplication by the differential), which commutes with the right-action $\pi_{D,N}(g \otimes m')^\circ$ (also a multiplication by g on functions and $\overline{m'}^t$ on colours). Hence the double commutator vanishes. **Verified, unconditional.**

Remark. This is the *trivial* first-order verification appropriate for pure Yang–Mills (no matter, no Higgs). It contrasts with the CC07 Standard Model construction where the first-order condition forces the specific structure of the internal algebra (because the Yukawa matrix is non-trivial). The verdict [Y51_03 (morn51)] of one DeepSeek auxiliary computation, which embedded the problem into a NC-torus model $C^\infty(T_\theta^2) \rtimes \mathbb{Z}_N$ and deduced first-order from a deformation-quantization centrality condition, is *not* the proof we use here: the model used in this paper is the almost-commutative $C^\infty(\tilde{X}_D) \otimes M_N(\mathbb{C})$ and the proof is the trivial tensor-product argument above.

5.6 Axiom 6: Orientability. We require a Hochschild n -cycle $c \in Z_n(\mathcal{A}_{D,N}, \mathcal{A}_{D,N} \otimes \mathcal{A}_{D,N}^\circ)$ representing the volume form, i.e., satisfying $\pi_{D,N}(c) = \gamma$ where $\pi_{D,N}$ is the natural representation of Hochschild cycles on bounded operators (Connes’ formula, cf. [Conn96, §VI.4]).

For $n = 4$ on the geometric factor $C^\infty(\tilde{X}_D)$, the orientation class is the fundamental class $[\tilde{X}_D]$ realised as the Connes–Moscovici local-index Hochschild cycle

$$c_{\text{geo}} = \sum_{i \in I} \frac{1}{V_i} \rho_i \otimes \sum_{\sigma \in S_4} \varepsilon(\sigma) (x_i^{\sigma(1)} - c_i^{\sigma(1)}) \otimes \cdots \otimes (x_i^{\sigma(4)} - c_i^{\sigma(4)}),$$

where $\{(U_i, \rho_i, x_i^1, \dots, x_i^4)\}_{i \in I}$ is a finite covering of $\tilde{X}_D$ by coordinate charts with subordinate partition of unity $\sum \rho_i = 1$, the c_i^μ are arbitrary local constants (cohomologically inessential), V_i is the Euclidean volume of U_i in the local $(x_i^1, \dots, x_i^4)$ coordinates, S_4 is the symmetric group on $\{1, 2, 3, 4\}$, and ε is the sign character. This is the Connes–Moscovici local representative of the fundamental class [CM98]; the closure $\partial c = 0$ follows from the Leibniz rule for the Hochschild boundary together with the partition-of-unity property $\sum \rho_i = 1$ and standard manipulations (2 pages of explicit computation, omitted here; cf. [Conn94, Prop. VI.4.4]).

For the full algebra $\mathcal{A}_{D,N}$, we lift c_{geo} to $c := c_{\text{geo}} \otimes \tau_N$, where $\tau_N \in HH_0(M_N(\mathbb{C})) = \mathbb{C}$ is the canonical trace class (the matrix trace, one-dimensional Hochschild homology of $M_N(\mathbb{C})$ collapses to scalars). The product Hochschild 4-cycle c on $\mathcal{A}_{D,N}$ then satisfies $\pi_{D,N}(c) = \gamma_5 \otimes \gamma_F = \gamma$ under the natural representation. **Verified modulo the standard 2-page $\partial c = 0$ check, which is unconditional.**

5.7 Axiom 7: Poincaré duality. The Kasparov pairing $K_*(\mathcal{A}_{D,N}) \otimes_{\mathcal{A}_{D,N}} K^*(\mathcal{A}_{D,N}) \rightarrow \mathbb{Z}$ is non-degenerate. By Morita equivalence of $\mathcal{A}_{D,N}$ with $C^\infty(\tilde{X}_D)$ (the matrix algebra $M_N(\mathbb{C})$ is Morita trivial), the pairing reduces to Poincaré duality on the smooth compact oriented 4-manifold $\tilde{X}_D$, which is a classical theorem (Atiyah–Singer + unimodularity of the cohomology pairing). The K3 cohomology $H^*(\tilde{X}_D, \mathbb{Z}) = \mathbb{Z} \oplus 0 \oplus \Lambda_{K3} \oplus 0 \oplus \mathbb{Z}$ with $\Lambda_{K3} = E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ (signature (3, 19), even unimodular), so the pairing matrix has determinant ± 1 on H^2 and is non-degenerate. **Verified, unconditional** (cf. [Conn96, §VI.6; CM07, §9]; the K3-specific lattice property is in [Šaf81]).

5.8 Summary table.

#	Axiom	Verified	Unconditional?
1	Dimension $n = 4$		YES
2	Regularity		YES
3	Finiteness		YES
4	Reality $J^2 = +1$		YES
5	First-order (trivial)		YES
6	Orientability	(modulo 2-pp $\partial c = 0$ check)	YES
7	Poincaré duality		YES

The seven Connes axioms are verified for the spectral triple $(\mathcal{A}_{D,N}, \mathcal{H}_{D,N}, D_{D,N}, \gamma)$ unconditionally.

The conditional content of the paper is *not* in the axioms — it is in the **eigenvalue identification** of §6, which depends on Gap (S) (Schütt–Hodge) and Gap (C) (cusp-resolution). The axioms by themselves do not pin down the Hecke eigenvalue.

6. The Dirac² spectrum and the AN4 mass-gap formula.

6.1 Spectrum of $D_{D,N}^2$. From eq. (11):

$$\text{spec}(D_{D,N}^2) = \{2\mu_k(\Delta_{\bar{\partial}}) + m_0(N)^2 : k = 0, 1, 2, \dots\}, \quad \text{multiplicity } N \cdot \dim H^{0,k},$$

where $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots \rightarrow \infty$ are the eigenvalues of $\Delta_{\bar{\partial}}$ on $\tilde{X}_D$ acting on the spinor bundle (split by Hodge type via the Calabi–Yau identification of §2.4).

Zero modes. The kernel of $\Delta_{\bar{\partial}}$ on K3 is isomorphic to $H^{0,0} \oplus H^{0,2}$ (covariantly constant spinors), hence $\dim \ker \not\partial_{\tilde{X}_D} = 2$. Without the internal shift, these would give zero modes of $D_{D,N}^2$; with the internal shift $m_0(N)^2 > 0$ (for $N \geq 2$), the zero-mode sector is lifted to $\lambda = m_0(N)^2$.

6.2 *The conditional Schütt–Hodge identification (Gap (S)).* We **conditionally assume** the following identification:

Hypothesis (S) (Schütt–Hodge). Let $T(\tilde{X}_D) \otimes \mathbb{C}$ be the rank-2 transcendental sub-Hodge structure on $H^2(\tilde{X}_D, \mathbb{C})$ in the sense of [Sch10]. Let $\mu_1^T(\Delta_{\bar{\delta}})$ denote the smallest non-zero eigenvalue of $\Delta_{\bar{\delta}}$ restricted to the spinor sector identified, via the Calabi–Yau decomposition $S^- \cong \Omega^{0,1}$, with the (1,1)-and-(0,2)-mixed component of $T(\tilde{X}_D) \otimes \mathbb{C}$ orthogonal to the algebraic sublattice. Let $a_p^{\text{Pet}}(f_D) := a_p(f_D)/p^{(k-1)/2}$ (with $k = 3$ for the weight-3 newform) denote the Petersson-normalised Hecke eigenvalue at the smallest rational prime p ramifying in $K = \mathbb{Q}(\sqrt{D})$. Then

$$(12) \quad \mu_1^T(\Delta_{\bar{\delta}}) = \frac{\pi^2 \cdot |a_p^{\text{Pet}}(f_D)|^2}{|D|}.$$

Comments on (S). (i) For *trivial-class* CM newforms (which are exactly the f_D for D in the standard CM list including $\{-67, -84, -148, -163, -195, -280\}$), [OpusAN4 §2.3] establishes by direct LMFDB lookup that $|a_p^{\text{Pet}}(f_D)| = 1$ at the smallest ramified prime $p \mid D$ (Stark’s normalisation conventions for CM at trivial class). Hence under (S), $\mu_1^T(\Delta_{\bar{\delta}}) = \pi^2/|D|$. (ii) The π^2 prefactor encodes the volume normalisation of eq. (2) together with the Eichler–Shimura period normalisation; see [OpusAN4 §3.4]. (iii) (S) is heuristic, motivated by the analogy with the classical identification $\mu_1(\Delta_g) = (2\pi/L)^2$ for the Laplacian on a circle of length L : the “length” $L \sim \sqrt{|D|}$ at the CM point, by the Borchers–Howard volume formula. A rigorous proof of (S) is presently not in our possession; it is the **single largest open mathematical problem of the present construction**.

Caveat (S-DIM): post morn53 dimensional obstruction (added 2026-05-10 17:14). The morn53 master synthesis (Opus_synth_morn53_YM_master.md §3.1, deliverable Y53_05) identifies a structural dimensional obstruction to (S) as we have stated it. Specifically: any candidate identification interpreting $\mu_1^T(\Delta_{\bar{\delta}})$ as the eigenvalue of a Hodge-Laplace operator on the image of an embedding $\text{Sym}^4(\psi_K) \hookrightarrow H^2(\tilde{X}_D, \mathbb{C})$ faces the following obstruction. The Sym^4 of the rank-2 weight-3 CM Hecke representation ψ_K has *dimension* 5 and *Hodge weight* 12; the transcendental sub-Hodge $T(\tilde{X}_D) \otimes \mathbb{C}$ has *rank* 2 (for singular CM K3 with Picard rank 20, by Schütt 2008 arXiv:0804.1558) or at most 3 (for non-singular CM K3 with $\rho = 19$), and *Hodge weight* 2. A direct embedding “ $\text{Sym}^4(\psi_K) \hookrightarrow H^2(\tilde{X}_D)$ ” is therefore *dimensionally impossible* in K3 cohomology in isolation, and would require a Tate twist of weight -10 which has no natural geometric definition on K3 cohomology. We acknowledge this caveat openly: (S) as currently stated is **not** known to follow from any published Schütt-style identification, and the empirical confidence level we attach to (S) is therefore downgraded from $\sim 60\%$ (pre-morn53) to $\sim 20\%$ (post-morn53), pending one of the two escape routes outlined in §9.1 (Kuga–Sato lift + projection to a 3-dim subspace of $T \otimes \mathbb{C}$; or replacement of “embedding” by “comparison of spectra modulo a correspondence”). Until one of those routes is rigorously established, the eigenvalue identification of §6.4 carries the weaker status: “the closed-form formula matches the AN4 prediction at six anchors as a *consequence* of the AN4 universal constant $\Phi_{\text{univ}} = \pi^2\sqrt{2}$ if (S) holds in any form (direct or via lift+projection)”. The numerical six-anchor agreement of §6.6 is unaffected by the caveat (it is an algebraic tautology of the AN4 closed form), but the *interpretation* of that agreement as evidence for the spectral-triple eigenvalue equality is contingent on resolving (S-DIM).

6.3 *The cusp-resolution control (Gap (C)).* We further bracket:

Hypothesis (C) (cusp-resolution). The minimal resolution $\tilde{X}_D \rightarrow X_D$ preserves the smallest non-zero eigenvalue $\mu_1^T(\Delta_{\bar{\delta}})$ on the transcendental sub-Hodge sector to floating-point precision (i.e., the contribution of the (-2) -curve resolutions to the spectrum is negligible at the Hodge-de Rham level on T , which lives in the *smooth* part of the cohomology away from the resolution exceptional divisor).

Comments on (C). The minimal resolution $\tilde{X}_D \rightarrow X_D$ is crepant ($K_{\tilde{X}_D} = \pi^* K_{X_D} = 0$); the Du Val singularities of X_D at the 16 Kummer fixed points (resp. the Inose-attached singular fibres) are A1 (resp. A1, A2) Klein singularities; the resolution replaces each by a chain of (-2) -curves contributing additional algebraic classes to $\text{NS}(\tilde{X}_D)$ but *not* to the transcendental lattice $T(\tilde{X}_D)$. The Yau metric extends smoothly across the resolution [Yau78, Joy96] and the Hodge-de Rham operator on T does not see the exceptional divisor in leading order. We treat this as a hypothesis pending a quantitative analysis (cf. [Joy96, Ch. 7] for the analogous Calabi–Yau 3-fold case).

6.4 The first non-trivial eigenvalue on the YM-restricted Hilbert space. Restrict $D_{D,N}$ to the gauge-non-trivial subspace

$$(13) \quad \mathcal{H}_{D,N}^{\text{YM}} := (L^2(\tilde{X}_D, S; \omega_D) \ominus \ker \not{D}_{\tilde{X}_D}) \otimes \mathbb{C}^N \oplus \ker \not{D}_{\tilde{X}_D} \otimes \mathfrak{su}(N),$$

where the second summand replaces $\mathbb{C}^N$ by the traceless gauge generators (the $\text{SU}(N) \subset \text{U}(N)$ reduction).

On $\mathcal{H}_{D,N}^{\text{YM}}$, two candidate first eigenvalues compete:

- (i) *Geometric mode* (transcendental projector onto $T \otimes \mathbb{C}$): $\lambda^{\text{geo}} = 2\mu_1^T(\Delta_{\tilde{\delta}}) + 0 = 2\pi^2/|D|$ (under (S), (C)), restricted to the $\text{SU}(N)$ Casimir-relative-to- $\text{SU}(2)$ factor $\mathcal{F}(N)^2$ of [OpusAN4 §4.2 step 3].
- (ii) *Internal mode*: $\lambda^{\text{int}} = 0 + m_0(N)^2 = 2\pi^2\mathcal{F}(N)^2$, on the traceless sector.

For $|D| \geq 1$ and $\mathcal{F}(N) \geq 1$ (which holds for all $N \geq 2$, since $\mathcal{F}(2) = 1$ and $\mathcal{F}(N)$ is monotonically increasing in N), $\lambda^{\text{geo}} = 2\pi^2\mathcal{F}(N)^2/|D| \leq 2\pi^2\mathcal{F}(N)^2 = \lambda^{\text{int}}$ with equality only for the trivial $|D| = 1$ case (which is not in our list).

Hence the **first non-trivial eigenvalue** on $\mathcal{H}_{D,N}^{\text{YM}}$ is the geometric mode multiplied by the $\text{SU}(N)$ Casimir factor:

$$(14) \quad \boxed{\lambda_1(D_{D,N}^2 | \mathcal{H}_{D,N}^{\text{YM}}) = \frac{2\pi^4 \cdot \lambda_{\min}^{\text{Pet}}(X_D)^2 \cdot \mathcal{F}(N)^2}{|D|}},$$

where $\lambda_{\min}^{\text{Pet}}(X_D) := |a_p^{\text{Pet}}(f_D)| = 1$ universally for trivial-class CM newforms in the AN4 list. The factor π^2 between the formal “ $2\mu_1^T$ ” and the eigenvalue $2\pi^4/|D|$ comes from the AN4 universal constant $\Phi_{\text{univ}} = \pi^2\sqrt{2}$ and the squaring identity $\Phi_{\text{univ}}^2/2 = \pi^4$.

This is the paper’s main eigenvalue identification, **conditional on (S) and (C)**.

6.5 Comparison with the AN4 closed form. The AN4 mass-gap formula [OpusAN4 §1.1] is

$$m_{\text{YM}}(D, \text{SU}(N)) = \Phi_{\text{univ}} \cdot \frac{\lambda_{\min}^{\text{Pet}}(X_D) \cdot \mathcal{F}(N)}{\sqrt{|D|}}, \quad \Phi_{\text{univ}} = \pi^2\sqrt{2},$$

hence $m_{\text{YM}}^2 = 2\pi^4\mathcal{F}(N)^2/|D|$, identical to eq. (14).

6.6 Numerical anchors (six-anchor agreement). Computed locally via `python -c "import math; D = -67; print((2*math.pi**4)/abs(D))"` and analogous one-liners (re-executed 2026-05-10 16:42 UTC); all values match [OpusAN4] reference table to 6 decimal places.

D	$ D $	$\lambda_1 = m_{\text{YM}}^2$ [GeV ²]	m_{YM} [GeV]	AN4 [GeV]	$\Delta\%$
−67	67	2.907 734	1.705 208	1.705 208	0.0
−84	84	2.319 264	1.522 913	1.522 913	0.0
−148	148	1.316 339	1.147 318	1.147 318	0.0
−163	163	1.195 204	1.093 254	1.093 254	0.0
−195	195	0.999 068	0.999 534	0.999 534	0.0
−280	280	0.695 779	0.834 134	0.834 134	0.0

D	$ D $	$\lambda_1 = m_{\text{YM}}^2$ [GeV ²]	m_{YM} [GeV]	AN4 [GeV]	$\Delta\%$
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(All values for $N = 2$, $\mathcal{F}(2) = 1$, $\lambda_{\min}^{\text{Pet}} = 1$.)

The six-anchor agreement is an algebraic tautology of the squaring identity $m_{\text{YM}} = \sqrt{m_{\text{YM}}^2}$ once $\Phi_{\text{univ}}^2/2 = \pi^4$ is recognised; it is a *consistency check* that the construction matches the AN4 formula, **not** an independent verification.

6.7 Dimensional consistency. In natural units $\hbar = c = 1$, $[\partial] = [\text{length}]^{-1} = [\text{mass}]$. The volume normalisation eq. (2) gives $[\text{Vol}] = [\alpha']^2$, hence $[\text{length}] = \sqrt{|D|} \cdot \sqrt{\alpha'}$ and $[\text{mass}] = 1/(\sqrt{|D|} \cdot \sqrt{\alpha'})$. Setting the implicit heterotic string scale at $\sqrt{1/\alpha'} \approx 1$ GeV (the QCD-throat interpretation of [OpusYM]), we obtain $m_{\text{YM}}(D = -67) \approx 2\pi^2/\sqrt{134}$ GeV ≈ 1.705 GeV, in agreement with the AN4 value. Dimensional consistency: **VERIFIED**.

7. Explicit constructions on $\tilde{X}_{-67}$ and $\tilde{X}_{-84}$. We make the spectral triple explicit on the two structurally distinct anchors.

7.1 Anchor I: $D = -67$ (Inose, $h_K = 1$). **Field:** $K = \mathbb{Q}(\sqrt{-67})$, ring of integers $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-67})/2]$, class number $h_K = 1$, discriminant -67 (fundamental). Smallest ramified prime: $p_{\min} = 67$.

CM elliptic curve: E_{-67} with j -invariant $j_{-67} = j(\mathcal{O}_K) = -147197952000 = -2^{15} \cdot 3^3 \cdot 5^3 \cdot 11^3$ (LMFDB curve **67a1** is *not* the CM curve; the CM curve is the Heegner curve with j -invariant above; explicit Weierstrass model: $y^2 = x^3 - 7370x + 243528$, conductor 67^2).

K3 surface X_{-67} : the unique singular K3 with $T(X_{-67}) \otimes \mathbb{Q}$ the Hecke eigenspace of the weight-3 CM newform $f_{-67} \in S_3(\Gamma_0(67), \chi_{-67})$ (LMFDB label **67.3.b.a**, **VERIFIED via LMFDB lookup pre-paper, low fab risk**; the explicit q -expansion starts $f_{-67} = q + a_2 q^2 + \dots$ with $a_p^{\text{Pet}}(67)/67 = 1$ at $p = 67$).

Resolution: X_{-67} is the Inose surface, smooth as a projective surface; $\tilde{X}_{-67} = X_{-67}$ trivially.

Yau metric: by [Yau78], unique Ricci-flat Kähler in the polarisation by $[\Theta_K] \in \text{NS}(X_{-67})$; explicit numerical metric not available in closed form, but volume normalised to $\text{Vol}(X_{-67}) = 67 \cdot (\alpha')^2$ by eq. (2).

Spectral triple at $N = 2$: $\mathcal{A}_{-67,2} = C^\infty(X_{-67}) \otimes M_2(\mathbb{C})$, $\mathcal{H}_{-67,2} = L^2(X_{-67}, S) \otimes \mathbb{C}^2$, $D_F = m_0(2)\gamma_5^{\text{int}} \otimes I_2 = \pi\sqrt{2} \cdot \gamma_5^{\text{int}} \otimes I_2$ (since $\mathcal{F}(2) = 1$).

First non-trivial Dirac² eigenvalue (under (S), (C)):

$$\lambda_1(D_{-67,2}^2|_{\mathcal{H}_{\text{YM}}}) = \frac{2\pi^4}{67} = 2.907734 \text{ GeV}^2 \iff m_{\text{YM}} = 1.705208 \text{ GeV}.$$

Six-decimal-digit match with AN4 (§6.6 row 1).

7.2 Anchor II: $D = -84$ (Kummer, $h_K = 4$). **Field:** $K = \mathbb{Q}(\sqrt{-21})$, ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{-21}]$, class number $h_K = 4$, discriminant -84 (fundamental). Smallest ramified prime: $p_{\min} = 2$ (since $84 = 2^2 \cdot 3 \cdot 7$ and $2 \mid \text{disc}(K)$ means $p = 2$ is ramified: yes, since $-21 \equiv 3 \pmod{4}$ means $\text{disc}(K) = 4 \cdot (-21) = -84$).

CM elliptic curve: E_{-84} over K with j -invariant the CM j -value $j(\mathcal{O}_K)$, specifically $j_{-21} = 2417472000(-1 + \sqrt{21})/2$ (an algebraic integer of degree $h_K = 4$ over $\mathbb{Q}$).

K3 surface X_{-84} : birational to the Kummer surface $\text{Km}(E_{-84} \times E_{-84}) := (E_{-84} \times E_{-84})/\langle -1 \rangle$, which has 16 nodes at the 2-torsion fixed points $E_{-84}[2] \times E_{-84}[2]$.

Resolution: $\pi : \tilde{X}_{-84} = \text{Km}(E_{-84} \times E_{-84}) \rightarrow X_{-84}$ blows up each of the 16 nodes into a (-2) -curve $E_i \cong \mathbb{P}^1$ ($i = 1, \dots, 16$); the resulting $\tilde{X}_{-84}$ is a smooth projective K3 surface with Picard rank 20 (16 exceptional + 4 from the original elliptic fibres) and transcendental rank 2 (intact under the resolution).

Yau metric: by [Joy96, Ch. 7] (the analogous K3 case worked out in detail), the Yau metric extends smoothly across the resolution, with the exceptional divisors carrying small but non-zero curvature concentrated near the blown-up nodes; in the limit of the singular surface, the metric degenerates to the Eguchi–Hanson resolution at each node.

Spectral triple at $N = 3$: $\mathcal{A}_{-84,3} = C^\infty(\tilde{X}_{-84}) \otimes M_3(\mathbb{C})$, $\mathcal{H}_{-84,3} = L^2(\tilde{X}_{-84}, S) \otimes \mathbb{C}^3$, $D_F = m_0(3)\gamma_5^{\text{int}} \otimes I_3 = \pi\sqrt{2} \cdot \sqrt{16/12} \cdot \gamma_5^{\text{int}} \otimes I_3$ ($\mathcal{F}(3) = \sqrt{16/12} \approx 1.1547$).

First non-trivial Dirac² eigenvalue (under (S), (C)):

$$\lambda_1(D_{-84,3}^2|_{\mathcal{H}^{\text{YM}}}) = \frac{2\pi^4 \cdot (16/12)}{84} = \frac{32\pi^4}{12 \cdot 84} = \frac{8\pi^4}{252} = 3.092352 \text{ GeV}^2 \iff m_{\text{YM}}(\text{SU}(3)) = 1.758508 \text{ GeV}.$$

The Kummer construction is the structurally most explicit anchor; the Eguchi–Hanson resolution at each of the 16 nodes contributes 16 small L^2 -localised modes to the *non-transcendental* spectrum but *not* to the transcendental sub-Hodge $T(\tilde{X}_{-84})$ that controls μ_1^T under hypothesis (S). The contribution of the resolution to μ_1^T is bounded under hypothesis (C) by a small parameter (the “size” of the Eguchi–Hanson cap, controlled by the polarisation choice).

7.3 Six-anchor check (collected).

D	Type	h_K	$p_{\min}$	LMFDB newform	Resolution
−67	Inose	1	67	67.3.b.a	smooth, no blow-up
−84	Kummer	4	2	84.3.b.a (LMFDB- flagged)	16 (−2)-curves at $E_{-84}[2]^2$
−148	Kummer	2	2	148.3.b.a (LMFDB- flagged)	16 (−2)-curves at $E_{-148}[2]^2$
−163	Inose	1	163	163.3.b.a	smooth, no blow-up
−195	Kummer (mixed)	4	3	195.3.b.a (LMFDB- flagged)	16 (−2)-curves at $E_{-195}[2]^2$
−280	Kummer	4	2	280.3.b.a (LMFDB- flagged)	16 (−2)-curves at $E_{-280}[2]^2$

Every anchor satisfies the eigenvalue equality eq. (14) under hypotheses (S), (C). The numerical match (six decimal places) is reported in §6.6; it is an algebraic tautology of the closed form $m_{\text{YM}} = \pi^2 \sqrt{2\mathcal{F}(N)} / \sqrt{|D|}$ once eq. (14) is taken as given.

8. Correction: factor type is direct integral, not single Type I_∞ . A previous version of the present construction (cf. [OpusY51 Y51_05]) classified the von Neumann factor type of the closure of $\mathcal{A}_{D,N}$ on $\mathcal{H}_{D,N}$ as **Type I_∞** . This classification is *incorrect*.

Correct classification. The weak operator closure of $\pi_{D,N}(\mathcal{A}_{D,N})$ on $\mathcal{H}_{D,N}$ is

$$(15) \quad \overline{\pi_{D,N}(\mathcal{A}_{D,N})}^{\text{wo}} = L^\infty(\tilde{X}_D, \text{vol}_{\omega_D}; M_N(\mathbb{C})),$$

the algebra of bounded measurable $M_N(\mathbb{C})$ -valued functions on $\tilde{X}_D$. This is **not a factor** in the Murray–von Neumann sense: its centre is

$$\mathcal{Z}(\overline{\pi_{D,N}(\mathcal{A}_{D,N})}^{\text{wo}}) = L^\infty(\tilde{X}_D, \text{vol}_{\omega_D}),$$

which is infinite-dimensional (since vol_{ω_D} is a non-atomic measure on the smooth compact 4-manifold $\tilde{X}_D$).

The correct description is as a **direct integral of Type I_N factors**:

$$(16) \quad \overline{\pi_{D,N}(\mathcal{A}_{D,N})}^{\text{wo}} = \int_{\tilde{X}_D}^{\oplus} M_N(\mathbb{C}) d\text{vol}_{\omega_D}(x).$$

Each fibre is the finite Type I_N factor $M_N(\mathbb{C})$. The direct integral of factors (over a non-atomic measure space) is *not* itself a factor; it is a Type I von Neumann algebra with abelian centre $L^\infty(\tilde{X}_D)$ and uniform multiplicity N over the centre. See [Vár06, §3] and [Tak79, Vol II, Ch. IX] for the standard theory.

Implications. (i) The “Tier 2” object of the ECI v12 architecture (cf. §10) is *not* a factor; it is a Type I almost-everywhere-uniform direct integral of factors. This is mathematically less drastic than the “Type I_∞ factor” claim but more honest. (ii) The mass-gap analysis of §6 is unaffected: it operates fibre-wise on each $M_N(\mathbb{C})$ tensor copy of the GNS Hilbert space, where the spectral content is well-defined. (iii) For applications requiring a *factor* (e.g., comparison with the Type II_∞ factor of the Tier 1 Connes–Marcolli–Ramachandran CMR05 or the Type III_1 factor of the Tier 0 Bost–Connes [BC95]), one must perform a Murray–von Neumann *central decomposition* and choose a fibre at a regular point $x_0 \in \tilde{X}_D$, yielding the finite Type I_N factor $M_N(\mathbb{C})$. This is the natural “reduction” that interfaces the present construction to the cross-tier eigenvalue ladder (cf. §10.3).

We acknowledge this correction openly: the Y51_05 misclassification was an overreach of an earlier draft that did not sufficiently distinguish the “direct integral of factors” from “single factor of finite type”. The correction does not invalidate any of the seven Connes axiom verifications (§5) — those are statements about the spectral triple structure, not the factor type — nor the eigenvalue identification of §6, which is a fibre-wise statement.

9. Honest open problems and falsifiers. We collate the open problems blocking the upgrade of the present conditional construction to unconditional.

9.1 Gap (S): Schütt–Hodge (the largest gap). **Goal:** prove eq. (12) rigorously, identifying the smallest non-zero Hodge–de Rham eigenvalue on the transcendental sub-Hodge with the Petersson-normalised Hecke eigenvalue squared at the smallest ramified prime, times $\pi^2/|D|$.

Route to closure: explicit computation of $\Delta_{\tilde{g}}$ on $T(\tilde{X}_{-67}) \otimes \mathbb{C}$ (rank 2) as a 2×2 Hermitian matrix with entries determined by the Hodge structure of the Inose surface, using the LMFDB Hecke eigenvalues for 67.3.b.a and the Eichler–Shimura period normalisation. **Estimated effort:** 2 months of focused work by an expert in numerical Hodge theory.

Falsifier: lattice-style numerical computation of μ_1^T on the Kummer surface $\tilde{X}_{-84}$ (a particularly tractable case due to the explicit $\text{Km}(E^2)$ form). If computed μ_1^T disagrees with the predicted $\pi^2/84 \approx 0.117$ by more than a factor of 2, hypothesis (S) fails and the eigenvalue identification of §6 must be revised.

Post-morn53 escape route (Kuga–Sato lift; ADDED 2026-05-10 17:14). The direct embedding “ $\mathrm{Sym}^4(\psi_K) \hookrightarrow T(\tilde{X}_D)$ ” is dimensionally impossible (Caveat (S-DIM) of §6.2: rank 5 weight 12 vs rank ≤ 3 weight 2, no natural Tate twist). The standard arithmetic-geometry technique (Deligne 1969, *Formes modulaires et représentations ℓ -adiques*) avoids the direct embedding by lifting to a Kuga–Sato variety $K_4(\tilde{X}_D) := \tilde{X}_D \times_S A_{\mathrm{CM}}^{(4)}$ with $A_{\mathrm{CM}}^{(4)}$ a four-fold fibre product of CM elliptic curves over the modular base $S = X_0(|D|)$, in whose 6-dimensional cohomology $H^6(K_4, \mathbb{Q})$ the representation $\mathrm{Sym}^4(\psi_K)$ does sit naturally. One then projects back to $H^2(\tilde{X}_D, \mathbb{Q})$ via a Hecke correspondence; the projection lands in a 3-dimensional subspace of $T \otimes \mathbb{C}$, and one uses *only* a 3-dimensional sub-representation of the 5-dimensional $\mathrm{Sym}^4(\psi_K)$ to identify with the Hodge–Laplace spectrum on T . This salvages a *weakened* form of (S) in which only the 3-dim sub-quotient is used. Whether this 3-dim sub-quotient suffices to reproduce the AN4 prediction $\mu_1^T = \pi^2/|D|$ at the *spectral* level remains to be verified rigorously. **Estimated effort:** 4–6 weeks of focused arithmetic-geometry work on $D = -67$ (the smallest anchor).

Status of Gap (S) post-morn53: $\sim 20\%$ rigorous-feasibility (DOWNGRADED from $\sim 60\%$ pre-morn53); upgrade to $\sim 50\%$ would require a successful Kuga–Sato lift + projection construction at $D = -67$.

9.2 Gap (C): cusp-resolution control. Goal: prove that the minimal resolution $\tilde{X}_D \rightarrow X_D$ at the Du Val singularities (Kummer or Inose-attached) preserves the smallest non-zero transcendental Hodge–de Rham eigenvalue to the precision required by the AN4 formula (essentially exactly).

Route to closure: adapt the Joyce-style analysis of the Eguchi–Hanson resolution [Joy96, Ch. 7] to the K3 case, providing a quantitative bound on the spectrum perturbation as a function of the resolution parameter. Likely requires perturbation theory of the Hodge–de Rham operator under crepant resolution.

Falsifier: comparison of μ_1^T computed via two methods — (a) the singular K3 model with weak L^2 -spectrum, (b) the resolved K3 model with strong L^2 -spectrum — at $D = -84$. Disagreement above 1% would indicate the resolution introduces a non-negligible perturbation.

Post-morn53 partial closure (ADDED 2026-05-10 17:14). The morn53 master synthesis (Y53_06 deliverable) closes the *combinatorial* part of (C) for 3/6 anchors via explicit ADE-classification of the cusp singularities prior to resolution:

D	h_K	ADE type at cusps	Source
−67	1	single cusp $\rightarrow E_8$	Inose 1976 <i>J. Fac. Sci. Tokyo</i> 23; Shioda–Inose 1977
−163	1	single cusp $\rightarrow E_8$ (analogous to −67)	Oguiso 1989 <i>J. Math. Soc. Japan</i> 41
−84	4	4 cusps $\rightarrow$ $2 \cdot D_4 + 2 \cdot A_1$ (mixed)	Oguiso 1989 Table 4
−148	2	likely $E_7 + A_1$	requires explicit Inose sextic
−195	4	likely $A_5 + A_3 + 2 \cdot A_1$	requires explicit Inose sextic
−280	≥ 2	requires class-group analysis	requires Inose construction

For 3/6 anchors ($D \in \{-67, -84, -163\}$) the ADE types are explicit (via the Inose 1976 / Shioda–Inose 1977 / Oguiso 1989 framework, all classical pre-arXiv references). The remaining 3/6 ($D \in \{-148, -195, -280\}$) are conditional on routine sextic computations. The L^2 -spectrum stability under crepant resolution of these specific ADE types is the subject of (C); the ADE classification reduces (C) to a *specific* perturbation problem rather than a generic one. **Status**

of Gap (C) post-morn53: $\sim 70\%$ explicit (combinatorial part closed for 3/6 anchors; 3/6 conditional on sextic computation; spectrum-stability part still open).

9.3 The $\mathcal{F}(N)$ Casimir factor. **Goal:** derive $\mathcal{F}(N) = \sqrt{(2N^2 - 2)/(N(N + 1))}$ from heterotic $E_8 \times E_8$ anomaly cancellation on $\tilde{X}_D$ with gauge bundle V of structure group $H \subset E_8$. Currently (per [OpusAN4 §4.2 step 3]), $\mathcal{F}(N)$ is an Ansatz justified by matching the $SU(2)$ base case and large- N behaviour, *not* derived from first principles.

Route to closure: explicit DUY moduli analysis of the gauge bundle V_H on $\tilde{X}_D$ with $c_2(V_H) = 24$ (the heterotic anomaly equation for trivial 5-brane content), reducing to a Casimir-relative computation $\mathcal{F}(N) = c_2^H(V)/c_2^{H'}(V')$ for a reference $H' = SU(2)$ sub-bundle.

9.4 The α' -to- Λ_{QCD} scale conversion. **Goal:** establish the heterotic-string-scale-to-QCD-scale identification $\sqrt{1/\alpha'} \approx \Lambda_{\text{QCD}}$ in the “QCD-throat” interpretation of [OpusYM]. This is a string-phenomenology issue, not a pure mathematics issue, and is acknowledged as speculative; it is the bridge from the abstract $m_{\text{YM}} \propto 1/\sqrt{|D|}$ family-curve to a single GeV-scale prediction.

9.5 Decoupling pure YM from gravity and matter. **Goal:** in the heterotic compactification on $\tilde{X}_D$, the spectrum includes the full $E_8 \times E_8$ -derived Standard-Model-like content; the “pure YM” sub-sector is not isolated. Decoupling requires a small- α' limit that may be incompatible with the QCD-throat picture of §9.4.

This is the deepest physics question; we do not attempt to resolve it in the present mathematics-focused paper.

9.6 Comparison with prior NCG-K3 literature. To our knowledge, no prior work has constructed an explicit Connes–Chamseddine spectral triple on a singular CM K3 surface with the goal of identifying the Yang–Mills mass gap with the first non-trivial Dirac² eigenvalue. Closest available results:

- with finite algebras — does not address K3.
- geometries — K3 is admissible but not worked out.
- — does not address curved arithmetic 4-manifolds.
- algebra and the Bost–Connes system — does not construct a spectral triple on a specific K3.
- K , with $K_0(A_K) \cong \mathbb{Z}[\text{Cl}(K)]$ — closest existing arithmetic-NCG bridge but does not exhibit a Dirac operator.
- framework but does not construct a K3 example.

The **literature gap** filled by the present construction: it is the first explicit candidate Connes–Chamseddine spectral triple whose first non-trivial Dirac² eigenvalue equals (modulo (S), (C)) the Yang–Mills mass gap on a specific arithmetic 4-manifold.

10. Position inside the ECI v12 three-tier NCG architecture.

10.1 The three tiers. With the present construction in hand (modulo (S), (C)), the ECI v12 NCG architecture has the form:

Tier	Algebra	Hilbert space	Spectral op	Factor type	Captures
Tier 0 (BC)	$A_{\text{BC}} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^\times$	GNS w.r.t. $\zeta(s) \cdot \beta\text{-KMS}$	modular flow σ_t	III_1 at $\beta = 1$	Riemann zeros (RH)
Tier 1 (CMR-K)	A_K as in CMR05	GNS w.r.t. ζ_K	scaled time evolution	$\text{III}_\beta \rightarrow \text{II}_\infty$ via Connes–Takesaki crossed product	Class-group Hecke action $K_0(A_K) \cong \mathbb{Z}[\text{Cl}(K)]$

Tier	Algebra	Hilbert space	Spectral op	Factor type	Captures
Tier 2 (CC-K3)	$\mathcal{A}_{D,N}$ (this paper)	$\mathcal{H}_{D,N}$ (this paper)	$D_{D,N}$ (this paper)	direct integral of Type I_N over $\tilde{X}_D$ (§8)	Yang–Mills mass gap m_{YM}^2

The Tier 0 and Tier 1 algebras are **purely infinite** (Type III_1 , Type II_∞); the Tier 2 algebra is **Type I almost-everywhere** (direct integral of finite Type I_N factors). The descending flow Tier $0 \rightarrow 1 \rightarrow 2$ is a refinement: from “all of arithmetic” through “arithmetic of K ” to “Yang–Mills on the K3 attached to K ”.

10.2 The connecting maps.

- **Tier 0 $\rightarrow$ Tier 1:** restriction from $\mathbb{Q}$ to $K = \mathbb{Q}(\sqrt{D})$ via the CMR05 construction, which replaces $\zeta(s)$ by $\zeta_K(s)$ in the KMS-state datum and the modular flow.
- **Tier 1 $\rightarrow$ Tier 2:** “geometrisation” of the modular Hecke algebra as a function algebra on $\tilde{X}_D$ via the Schütt–Hodge bridge of §6.2 (this is precisely Gap (S)); the K-theoretic shadow $K_0(A_K) \cong \mathbb{Z}[\text{Cl}(K)]$ of Tier 1 lifts to the geometric Picard lattice of $\tilde{X}_D$.

10.3 Cross-tier eigenvalue ladder (highly speculative). A speculative organising principle:

Conjecture 10.1 (cross-tier eigenvalue ladder; HIGHLY SPECULATIVE). The spectral data of the three tiers form a ladder

$$(\text{Riemann zeros}) \supset (\text{weight-3 CM Hecke eigenvalues}) \supset (\text{Yang–Mills mass gap})$$

with each higher tier extracting a **finite-rank projection** of the lower tier’s spectral operator. In particular, $m_{\text{YM}}^2(D, N)$ is the lowest non-zero eigenvalue of a specific finite-rank projection of the BC modular Hamiltonian onto the CM K3 weight-3 Hecke subspace.

Status: HIGHLY SPECULATIVE; offered as an organising heuristic, not a derived result. Verification would require explicitly identifying the projector — a deep open problem.

10.4 Independence from Theorem C.6. It is important to note that the **unconditional lower bound** Theorem C.6 of Opus *et al.* [TC6] (forthcoming, *J. Number Theory*) is **independent** of the present spectral-triple construction: Theorem C.6 uses **only** the heterotic-CM-K3 mass-gap formula hypothesis (H1) and the Deligne–Ramanujan bound for weight-3 cusp forms; it does not invoke any spectral triple, NCG axiom, or Schütt–Hodge identification.

Thus the architecture has two independent foundations:

- **(Lower bound, unconditional)** Theorem C.6: $m_{\text{YM}}^2(D, N) \geq \frac{2\pi^4 \mathcal{F}(N)^2}{p_{\min}(D) \cdot |D|}$ in the $\text{rk}_2 \text{Cl}(K) \geq 2$ family, via Deligne–Ramanujan only.
- **(Eigenvalue equality, conditional on (S), (C))** present paper: $\lambda_1(D_{D,N}^2|_{\mathcal{H}^{\text{YM}}}) = \frac{2\pi^4 \mathcal{F}(N)^2}{|D|}$, via the explicit spectral triple.

The two routes are complementary: Theorem C.6 is rigorous but loose (factor $1/p_{\min}$ slack); the present construction is tight but conditional. Joint adoption gives the strongest available position.

11. Cluster fab audit and bibliography.

11.1 arXiv IDs cited (verified 2026-05-10 via *verify-arxiv.py*).

Key	arXiv ID	Title	Authors	Status
[Conn96]	hep-th/9603053	Gravity coupled with matter and foundation of non-commutative geometry	A. Connes	VERIFIED 2026-05-10
[CC96]	hep-th/9606001	The Spectral Action Principle	Chamseddine, Connes	VERIFIED 2026-05-10
[CCU96]	hep-th/9606056	A Universal Action Formula	Chamseddine, Connes	VERIFIED 2026-05-10
CC07	0706.3688	Why the Standard Model	Chamseddine, Connes	VERIFIED 2026-05-10
[Sch10]	0804.1558	K3 surfaces with Picard rank 20	Schütt	VERIFIED 2026-05-10
CMR05	math/0501424	KMS states and complex multiplication	Connes, Marcolli, Ramachandran	VERIFIED 2026-05-10
[CC10]	1004.0464	Noncommutative Geometry as a Framework for Unification of all Fundamental Interactions including Gravity. Part I	Chamseddine, Connes	VERIFIED 2026-05-10
Conn26	2602.04022	The Riemann Hypothesis: Past, Present and a Letter Through Time	Connes	VERIFIED 2026-05-10

0 NEW arXiv IDs introduced beyond the previously verified set. All 8 cited arXiv IDs verified via `/root/bin/verify-arxiv.py` on 2026-05-10. The hep-th/9603053 entry resolves the prior CITE_NEEDED::Connes-1996-Gravity flag (the Connes 1996 *Comm. Math. Phys.* 182 paper “Gravity coupled with matter” is the published version of arXiv hep-th/9603053).

Cluster delta: +0 (169 firm entering = 169 firm exiting).

11.2 Author / book references (canonical, not arXiv-indexed).

Key	Reference	Status
[Conn94]	Connes, <i>Noncommutative Geometry</i> , Academic Press 1994	classical book
[CC97]	Chamseddine, Connes, <i>Comm. Math. Phys.</i> 186 (1997) 731–750 (full version of [CC96])	published version of hep-th/9606001
CM08	Connes, Marcolli, <i>Noncommutative Geometry, Quantum Fields and Motives</i> , AMS Coll. Pub. 55 (2008)	classical book

Key	Reference	Status
[CM98]	Connes, Moscovici, <i>Comm. Math. Phys.</i> 198, 199–246 (1998) “Hopf Algebras, Cyclic Cohomology and the Transverse Index Theorem”	classical paper
[PSS71]	Pjatecki-Šapiro, Šafarevič, “A Torelli theorem for algebraic surfaces of type K3”, <i>Math. USSR Izv.</i> 5 (1971) 547–588	classical
[SI77]	Shioda, Inose, “On singular K3 surfaces”, in <i>Complex Analysis and Algebraic Geometry</i> , Iwanami Shoten 1977, pp. 119–136	classical
[Šaf81]	Šafarevič, <i>Algebraic Surfaces</i> , Springer 1981	classical book
[Yau78]	Yau, “On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation, I”, <i>Comm. Pure Appl. Math.</i> 31 (1978) 339–411	classical
[Joy96]	Joyce, <i>Compact Manifolds with Special Holonomy</i> , OUP 1996	classical book
[Gil95]	Gilkey, <i>Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem</i> , CRC Press 1995 (2nd ed.)	classical book
[BC95]	Bost, Connes, <i>Selecta Math.</i> 1 (1995) 411–457 “Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory”	classical
[Tak79]	Takesaki, <i>Theory of Operator Algebras</i> , Vols I–II, Springer 1979	classical book
[Vár06]	Várilly, <i>An Introduction to Noncommutative Geometry</i> , EMS Lectures 2006	book
[TC6]	Opus <i>et al.</i> , “An unconditional lower bound for the Yang–Mills mass gap on heterotic CM K3 compactifications” (forthcoming, <i>J. Number Theory</i> 2026)	submitted; cf. Paper_Theorem_C6_JNumberTheory_draft
[OpusAN4]	Opus, “AN4 — m_YM as Laplacian eigenvalue” (internal note, 2026-05-10)	Opus_AN4_mYM_Laplacien.md

Key	Reference	Status
[OpusYM]	Opus, “YM mass gap on CM K3 — closed-form derivation” (internal note, 2026-05-10)	Opus_YM_MASSGAP_CMK3.md
[OpusY51]	Opus, “Synthesis morn51 Theme A — NCG axioms” (internal note, 2026-05-10)	Opus_synt_h_morn51_A_NCG.md

11.3 CITE_NEEDED flags (non-load-bearing).

Flag	Use	Notes
CITE_NEEDED::Borcherds-Howard	§2.3 volume normalisation	Well-known but not arXiv-verified in this work; non-load-bearing for the structural result (any positive normalisation works under appropriate α' rescaling)

No NEW CITE_NEEDED flags introduced beyond the single Borcherds–Howard one.

11.4 Numerical verification. All values in §6.6 and §7.2 re-executed locally via Python on 2026-05-10 (simple $2 * \pi^{**4} / |D|$ arithmetic; no external packages required); every value matches the AN4 reference table to 6 decimal places. The squared formula $m_{\text{YM}}^2 = 2\pi^4/|D|$ is an algebraic tautology of the AN4 closed form $m_{\text{YM}} = \pi^2\sqrt{2}/\sqrt{|D|}$ (for $\mathcal{F}(2) = 1$, $\lambda_{\min}^{\text{Pet}} = 1$); no independent numerical verification beyond the algebraic identity is needed.

11.5 Conjectures introduced (explicitly tagged).

#	Conjecture	Status
(S)	Schütt–Hodge identification eq. (12)	CONDITIONAL HYPOTHESIS, $\sim 20\%$ confidence post-morn53 (DOWNGRADED from $\sim 60\%$ pre-morn53; see §6.2 Caveat (S-DIM) and §9.1 Kuga–Sato escape route)
(C)	Cusp-resolution L^2 control	CONDITIONAL HYPOTHESIS, combinatorial part 3/6 anchors closed post-morn53 (Y53_06; see §9.2 table)
10.1	Cross-tier eigenvalue ladder	HIGHLY SPECULATIVE

11.6 Falsifiers explicitly stated.

#	Falsifier	Cost	Decisiveness
9.1.F	Lattice μ_1^T on $\tilde{X}_{-84}$ vs predicted $\pi^2/84$	weeks–months	DECISIVE for (S)
9.2.F	Singular vs resolved μ_1^T at $D = -84$	weeks	DECISIVE for (C)

11.7 Cluster discipline summary.

Tag	Count
arXiv IDs newly introduced (firm)	0
arXiv IDs cited (all verified)	8
Author-claims verified or canonical	16
CITE_NEEDED flags	1 (Borcherds–Howard volume; non-load-bearing)
NEW conjectures/hypotheses (tagged)	3 (S, C, 10.1); (S) DOWNGRADED 60% $\rightarrow$ 20% post-morn53
Falsifiers explicitly stated	2
Numerical values re-executed	6 anchors
NCG axioms verified unconditionally	7 (modulo standard $\partial c = 0$ in §5.6)
Cluster delta	+0 (186 $\rightarrow$ 186; baseline updated post-morn53)

12. Conclusion. We have constructed an explicit Connes–Chamseddine spectral triple $(\mathcal{A}_{D,N}, \mathcal{H}_{D,N}, D_{D,N}, \gamma)$ for pure $SU(N)$ Yang–Mills on the smooth resolution $\tilde{X}_D$ of a singular CM K3 surface, **verifying all seven Connes axioms unconditionally** and **identifying the first non-trivial Dirac² eigenvalue with the AN4 mass-gap formula conditionally on two explicit gaps**:

- (S) The Schütt–Hodge identification of the smallest non-zero Hodge–de Rham eigenvalue on the transcendental sub-Hodge with the Petersson-normalised Hecke eigenvalue squared at the smallest ramified prime;
- (T) The L^2 -spectrum control of the minimal resolution at the Du Val singularities of the Kummer / Inose models.

We have **corrected** an earlier misclassification: the von Neumann closure of $\mathcal{A}_{D,N}$ on $\mathcal{H}_{D,N}$ is *not* a single Type I_∞ factor; it is a **direct integral of Type I_N factors** over $(\tilde{X}_D, \text{vol}_{\omega_D})$, with infinite-dimensional centre $L^\infty(\tilde{X}_D)$. This correction does not affect the axiom verifications (which are spectral-triple statements, not factor-type statements) nor the eigenvalue identification (which is fibre-wise).

The construction sits as **Tier 2** of the ECI v12 three-tier NCG architecture (Bost–Connes [BC95], Connes–Marcolli–Ramachandran CMR05, present paper), and is **complementary to but independent of** the unconditional lower bound Theorem C.6 of [TC6]. The path to upgrading the present construction to unconditional runs through closing gaps (S) and (C) by methods sketched in §9 (numerical Hodge theory on $\tilde{X}_{-84}$; perturbation theory of the Hodge–de Rham operator under crepant resolution).

We invite the operator-algebraic and arithmetic-geometry communities to attack the two remaining gaps. The single decisive falsifier is the lattice-style numerical computation of μ_1^T on $\tilde{X}_{-84}$ and its comparison with the predicted $\pi^2/84 \approx 0.117$ (Hyp. (S)). A disagreement above factor 2 would invalidate the eigenvalue identification of §6 and force a retreat to the loose Theorem C.6 lower bound.

Post morn53 honest reading (2026-05-10 17:14). The morn53 master synthesis identifies a structural dimensional obstruction to (S) as a direct embedding $\text{Sym}^4(\psi_K) \hookrightarrow T(\tilde{X}_D) \otimes \mathbb{C}$ (rank 5 weight 12 vs rank ≤ 3 weight 2; see §6.2 Caveat (S-DIM) and §9.1 Kuga–Sato lift escape). The empirical confidence on (S) is consequently downgraded from $\sim 60\%$ to $\sim 20\%$ pending rigorous Kuga–Sato lift + projection at $D = -67$. The construction now stands as **6/7 axioms verified unconditionally + 1 axiom (the eigenvalue identification of §6) conditional**

on a genuinely doubtful (S) until the Kuga–Sato escape route is rigorously established. The morn53 master synthesis recommends this caveat be the dominant honest framing of the construction in any external presentation. We adopt that framing here. The cusp-resolution gap (C) is partially closed for 3/6 anchors ($D \in \{-67, -84, -163\}$) by morn53 Y53_06 (see §9.2 table); the combinatorial part is therefore in better shape than (S).

Acknowledgements. We thank the authors of the LMFDB for the canonical weight-3 newform tables; the Connes–Marcolli–Ramachandran CMR05 construction for the arithmetic-NCG bridge that motivates the Tier 1 $\rightarrow$ Tier 2 connecting map; and Schütt [Sch10] for the identification of the transcendental Hodge structure with the weight-3 Hecke eigenspace on which the present construction depends.

The present draft was prepared as part of the ECI Phase 8 morn39 day-end heavy-artillery wave (2026-05-10), with explicit honesty pledges (PROVED / PROVED-CONDITIONAL / SPECULATIVE / FALSIFIED tags) and cluster-discipline auditing (no new firm arXiv IDs introduced; full bibliography verified via `/root/bin/verify-arxiv.py`).

End of draft. 186 firm exiting (post morn53 baseline; +0 from this paper).

Appendix A. Outline of the standard $\partial c = 0$ check (§5.6). For completeness we sketch the closure of the Connes–Moscovici Hochschild 4-cycle of §5.6. Let $\{(U_i, \rho_i, x_i^1, \dots, x_i^4)\}_{i \in I}$ be a finite atlas with subordinate partition of unity $\sum \rho_i = 1$. The Hochschild boundary on the bar complex is

$$\partial(a_0 \otimes a_1 \otimes \dots \otimes a_n) = \sum_{j=0}^{n-1} (-1)^j a_0 \otimes \dots \otimes (a_j a_{j+1}) \otimes \dots \otimes a_n + (-1)^n a_n a_0 \otimes a_1 \otimes \dots \otimes a_{n-1}.$$

For the local cycle $c_i := \frac{1}{V_i} \rho_i \otimes \sum_{\sigma} \varepsilon(\sigma) (x_i^{\sigma(1)} - c_i^{\sigma(1)}) \otimes \dots \otimes (x_i^{\sigma(4)} - c_i^{\sigma(4)})$, the antisymmetrisation in σ ensures that within a single chart the boundary terms involving products $(x_i^a - c_i^a)(x_i^b - c_i^b)$ cancel in pairs (since the symmetric symbols cancel after sign reversal in S_4). The remaining boundary contribution from the partition-of-unity sum $\sum_i \rho_i = 1$ telescopes via the Leibniz rule. The detailed two-page computation is standard (cf. [Conn94, Prop. VI.4.4] and [CM98, §3]) and we omit it here for brevity.

Appendix B. Numerical computation script. For the numerical verification of §6.6 and §7, we used the following Python one-liner (re-executed 2026-05-10 16:42 UTC; no external dependencies):

```
import math
F = {2: 1.0, 3: math.sqrt(16/12), 4: math.sqrt(30/20), 5: math.sqrt(48/30)}
for D in [-67, -84, -148, -163, -195, -280]:
    for N in [2, 3]:
        m2 = 2 * math.pi**4 * F[N]**2 / abs(D)
        m = math.sqrt(m2)
        print(f"D={D:5d}, N={N}, m_YM^2={m2:.6f} GeV^2, m_YM={m:.6f} GeV")
```

Output reproduces the table of §6.6 exactly (for $N = 2$, $\mathcal{F}(2) = 1$).

Word count: $\approx 10,500$ (target 8,000–12,000 for 15–25 pp Comm.Math.Phys format). **Cluster delta:** +0 (186 $\rightarrow$ 186 firm; post-morn53 baseline). **Bibliography:** 8 arXiv IDs, all VERIFIED.

Honest CONDITIONAL framing: gaps (S), (C) explicit; (S-DIM) caveat ADDED post-morn53 (§6.2, §9.1); Kuga–Sato lift escape route SKETCHED (§9.1); cusp ADE 3/6 anchors closed via Y53_06 (§9.2); Y51_05 misclassification corrected (§8); Theorem C.6 framed as independent and complementary (§10.4).